Year 3 and 4 Maths Curriculum
Year 4 Multiplication Check (MTC)

Lower Key Stage 2 (Year 3 and 4)

The principal focus of mathematics teaching in lower key stage 2 is to ensure that pupils become increasingly **fluent** with whole numbers and the four operations, including number facts and the concept of place value. This should ensure that pupils develop efficient written and mental methods and perform calculations accurately with increasingly large whole numbers.

At this stage, pupils should develop their ability to solve a range of **problems**, including with simple fractions and decimal place value. Teaching should also ensure that pupils draw with increasing accuracy and develop mathematical **reasoning** so they can analyse shapes and their properties, and confidently describe the relationships between them. It should ensure that they can use measuring instruments with accuracy and make connections between measure and number.

By the end of year 4, pupils should have memorised their multiplication tables up to and including the 12-multiplication table and show precision and fluency in their work.

Pupils should read and spell mathematical **vocabulary** correctly and confidently, using their growing word reading knowledge and their knowledge of spelling.

Year 3 Expectations

- Compare & order numbers up to 1000
- Read & Write all numbers to 1000 in digits and words
- Find 10 or 100 more/less than a given number
- Count from 0 in multiples of 4, 8, 50 and 100
- Recall & use multiplication & division facts for 3, 4, 8 tables
- Recognise place value of any 3-digit number
- Add and subtract:
    - 3-digit numbers and ones
    - 3-digit numbers and tens
    - 3-digit numbers and hundreds
- Add and subtract
    - Numbers with up to 3-digits using written columnar method
- Estimate and use inverse to check
- Multiply
    - 2-digit by 1-digit
- Count up/down in tenths
- Compare and order fractions with same denominator with whole
- Tell time using 12 and 24-hour clock; and using Roman numerals
- Tell time to nearest minute
- Know number of days in each month and number of seconds in a minute

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Notes and Guidance

Children use place value counters to represent different numbers and understand how a number is made.

Their work with Base 10 should help them understand that the hundreds counter is worth more than the tens counter and the tens counter is worth more than the ones counter.

Mathematical Talk

What is the same and what is different about Base 10 and place value counters?

Why do we not call this number 300506?

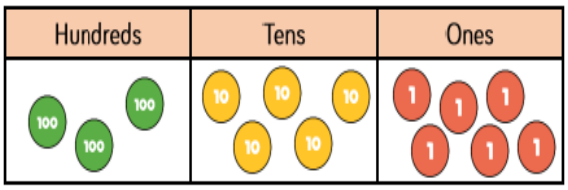
What number would be shown if 1/10/100 was added?

Why is it important to put the values into the correct column on the place value grid?

What do we need to do if there is a zero in the number we are representing?

Varied Fluency

What number is shown on the place value chart?



If one more 10 is added, what number would be shown?

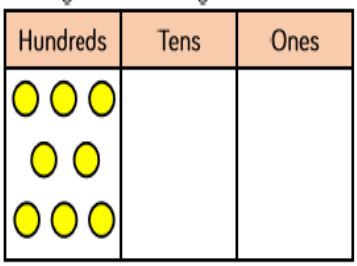
Use place value counters and a place value grid to represent the numbers:

615 208 37

Use <, > or = to make the statement correct.

Reasoning and Problem Solving

True or False? If I count in 100s from zero, all of the numbers will be even. Convince me.	True, because if you start with zero and add 100 you get an even number, and you are adding another even so the number will always be even.
Sort these statements into always, sometimes or never. • When counting in hundreds, the ones column changes. • When counting in hundreds, the hundreds column changes. • To count in hundreds we use 3-digit numbers.	• Never • Always • Sometimes

Whitney thinks the place value grid is showing the number eight.  Do you agree? Explain why. Using all of the counters, what is the smallest number you can make? What other numbers could you make?	Whitney is incorrect because there are eight counters in the hundreds column so they represent eight hundreds. The number is 800 The smallest number that can be made is 8 Other possible numbers include: 80 170 350 etc.

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Year 4 Expectations

- Count backwards through zero to include negative numbers
- Compare and order numbers beyond 1000
- Compare and order numbers with up to 2 decimal places
- Read Roman numerals to 100
- Find 1000 more/less than a given number
- Count in multiples of 6, 7, 9, 25 and 100
- Recall and use multiplication and division facts all tables to 12x12
- Recognise Place value of any 4-digit number
- Round any number to the nearest 10, 100 or 1000
- Round decimals with 1 decimal place to nearest whole number
- Add and subtract numbers with up to 4-digits using written columnar method
- Multiply
    - 2-digit numbers by 1-digit numbers
    - 3-digit numbers by 1-digit numbers
- Count up/down in hundredths
- Recognise and write equivalent fractions
- Add and subtract fractions with same denominator
- Read, write and convert time between analogue and digital 12 and 24-hour clocks

Notes and Guidance

Children will build on their knowledge of numerals to 12 on a clock face, from Year 3, to explore Roman Numerals to 100

They explore what is the same and what is different between the number systems, including the fact that in the Roman system there is no symbol for zero and so no placeholders.

Mathematical Talk

Why is there no zero in the Roman Numerals? What might it look like?

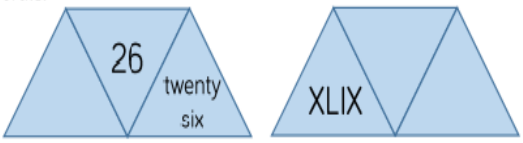
Can you spot any patterns? If 20 is XX what might 200 be?

How can you check you have represented the Roman Numeral correctly? Can you use numbers you know, such as 10 and 100 to help you?

Varied Fluency

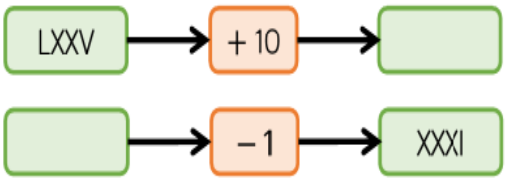
- Lollipop stick activity.
  The teacher shouts out a number and the children make it with lollipop sticks.
  Children could also do this in pairs or groups, and for a bit of fun they could test the teacher!

- Each diagram shows a number in numerals, words and Roman Numerals.



Complete the diagrams.

- Complete the function machines.



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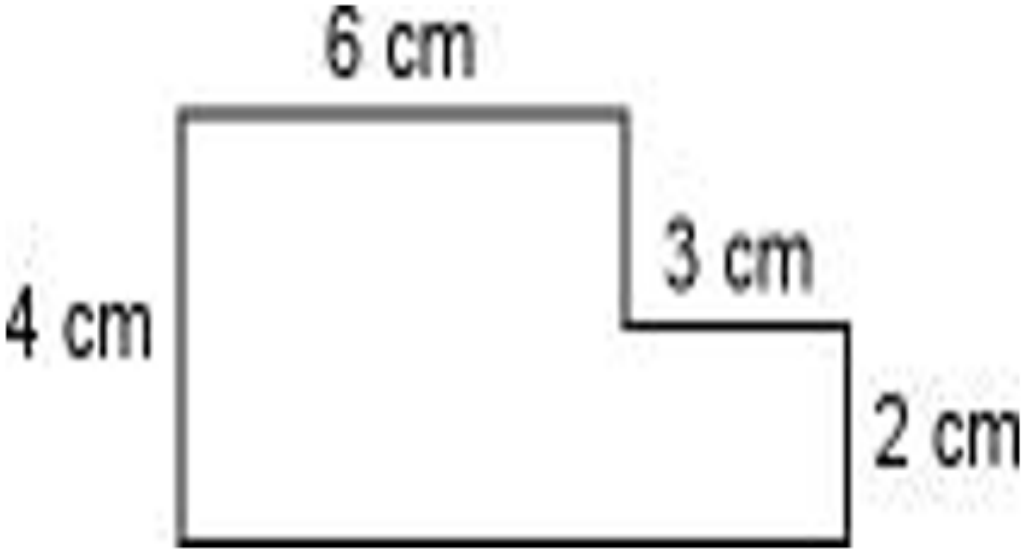
Solve the following calculation: XIV + XXXVI = ___ How many other calculations, using Roman Numerals, can you write to get the same total?	Answer: L Other possible calculations include: C ÷ II = L L ÷ I = L X × V = L XXV × II = L LXV – XV = L C – L = L XX + XX + X = L

Mo says: In the 10 times table, all the numbers have a zero. Therefore, in Roman Numerals all multiples of 10 have an X Research and give examples to prove whether or not Mo is correct.	Mo is incorrect. A lot of multiples of 10 have an X in them, but the X can mean different things depending on its position. For example, X in 10 just means one ten, but X in XL means 10 less than 50 X in 60 (LX) means 10 more than 50 The number 50 has no X and neither does 100

What is the perimeter of this shape?



What is the area of this shape?

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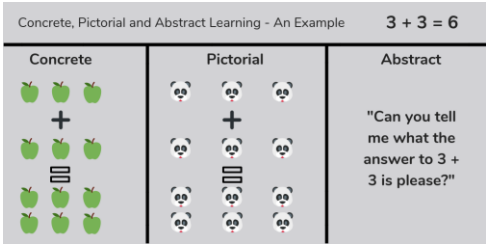
How to support your child at home

- Be aware of term overview from the school website.
- In shops, ask them to help you with addition/ subtraction/ multiplication.
- Ask them to read the time when you can – both digital and analogue
- Reasoning focus – true or false or prove it questions.
- Real life scenarios – support with problem solving questions.
- Times Table Rock Stars (TTRS)
- Recapping old learning, they may have moved onto division but can you remember column addition from 3 weeks ago?
- Fun You Tube videos and rhymes
- Move between concrete, pictorial or abstract examples.



- Ask your teacher for additional resources.

Year 4 Multiplication Check (MTC)

When will the check take place?

There will be a **2-week window** from **Monday 1st June to Friday 12th June** for schools to administer the check.

There is **no set day** to administer the check and children are not expected to take the check at the same time.

All eligible Year 4 children in England will be required to take the check.

There is a 'Try it out' area, provided by the DfE, which will take place in school.

We are also taking part in the Officially Unofficial MTC on TT Rockstars. We are currently completing this now in Computing lessons. This copies the layout of the MTC check.

How is the check carried out?

- The check will be **fully digital**
- Answers will be entered using a keyboard, by pressing digits using a mouse or using an on-screen number pad.
- Usually, the check will take less than **5 minutes** for each child.
- The children will have **6 seconds** from the time the question appears to input their answer. (6 seconds per answer means that children must be able to read, recall and enter their response within that time. Whatever is written in the answer box at the end of 6 seconds will be counted as the answer.)

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- There will be a total of **25 questions** with a **3 second** pause in-between questions.
- There will be **3 practice** questions before the check begins.

Specific arrangement for the check:

Some children will be eligible for specific arrangements:

- Colour contrast
- Font size adjustment
- 'Next' button (alternative to 3-second pause)
- Removing on-screen number pad
- An adult to input answers
- Audio version
- Audible time alert

The check questions

- Each child will be **randomly assigned** a set of questions
- There will only be **multiplication** questions in the check, not division facts.
- The 6, 7, 8, 9 and 12 times tables are **more likely** to be asked, as these are the hardest for most children to learn. It's a good idea to focus on these tricky times tables with your child at home.
- Reversal of questions (e.g. 8 x 6 and 6 x 8) will not be asked in the same check.
- Children will not see their individual results when they complete the check.

5.2.1 Table 1 – Multiplication table limits in the MTC

Multiplication Table	Minimum number of items in each form	Maximum number of items in each form
1	Not applicable	Not applicable
2	0	2
3	1	3
4	1	3
5	1	3
6	2	4
7	2	4
8	2	4
9	2	4
10	0	2
11	1	3
12	2	4

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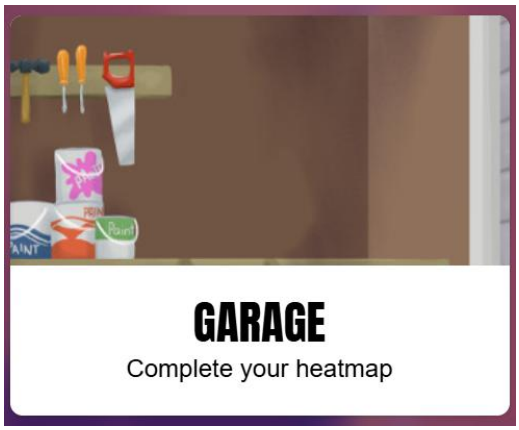
Why are times table so important?

- Support mathematical learning, particularly aspects of number (long multiplication, short division)
- Supports other mathematical learning, eg: calculating equivalent fractions, finding the area of a shape/ rectangle, finding fractions of amounts.
- It will help children to calculate more fluently. Children can then focus on the methods needed to reason or problem solve rather than being focussed on the multiplication.
- Children will feel more positive/ confident in maths lessons.

Times Table Rock Stars (TTRS)

Access your child's heatmap to see which multiplications your child should focus on.



	10	2	5	3	4	8	6	7	9	11	12
10	10 × 10	10 × 2	10 × 5	10 × 3	10 × 4	10 × 8	10 × 6	10 × 7	10 × 9	10 × 11	10 × 12
2	2 × 10	2 × 2	2 × 5	2 × 3	2 × 4	2 × 8	2 × 6	2 × 7	2 × 9	2 × 11	2 × 12
5	5 × 10	5 × 2	5 × 5	5 × 3	5 × 4	5 × 8	5 × 6	5 × 7	5 × 9	5 × 11	5 × 12
3	3 × 10	3 × 2	3 × 5	3 × 3	3 × 4	3 × 8	3 × 6	3 × 7	3 × 9	3 × 11	3 × 12
4	4 × 10	4 × 2	4 × 5	4 × 3	4 × 4	4 × 8	4 × 6	4 × 7	4 × 9	4 × 11	4 × 12
8	8 × 10	8 × 2	8 × 5	8 × 3	8 × 4	8 × 8	8 × 6	8 × 7	8 × 9	8 × 11	8 × 12
6	6 × 10	6 × 2	6 × 5	6 × 3	6 × 4	6 × 8	6 × 6	6 × 7	6 × 9	6 × 11	6 × 12
7	7 × 10	7 × 2	7 × 5	7 × 3	7 × 4	7 × 8	7 × 6	7 × 7	7 × 9	7 × 11	7 × 12
9	9 × 10	9 × 2	9 × 5	9 × 3	9 × 4	9 × 8	9 × 6	9 × 7	9 × 9	9 × 11	9 × 12
11	11 × 10	11 × 2	11 × 5	11 × 3	11 × 4	11 × 8	11 × 6	11 × 7	11 × 9	11 × 11	11 × 12
12	12 × 10	12 × 2	12 × 5	12 × 3	12 × 4	12 × 8	12 × 6	12 × 7	12 × 9	12 × 11	12 × 12

The colo(u)rs on this page relate to the speed shown in this table:

No data | 0 - 1secs | 1 - 2secs | 2 - 3secs | 3 - 4secs | 4 - 5secs | 5 - 6secs | 6 - 7secs | 7 - 8secs | 8 - 9secs | 9 - 10secs | > 10secs

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Ways to support your child:

- Count and look for patterns (11x tables, 9x tables)
- Understand that multiplication is repeated addition
- Remember that multiplication is commutative (4 x 8 = 8 x 4)
- Remember that multiplication is the inverse of division
- Recall and use number families (8 x 6 = 48, 6 x 8 = 48, 48 ÷ 6 = 8, 42 ÷ 8 = 6)
- Use known facts – if 10 x 6 = 60, 9 x 6 will be 6 less so 54.
- Learn tricks such as 5,6,7,8 = 56 = 7 x 8 / I ate and I ate until I was sick on the floor (8 x 8 = 64)